UNIVERSITY OF TORONTO
Faculty of Arts & Science

April 2024 Examinations

CSC 311 H1S

Duration: 3 hours

Aids Allowed: One *double*-sided *hand*written *8.5"×11"* aid sheet and a non-programmable calculator.

*Do **not** turn this page until you have received the signal to start.*
In the meantime, write your name and student number below (please do this now!) and *carefully* read *all* the information on the rest of this page.

First name (please write as legibly as possible within the boxes)

Last name

Student Number

- Turn off and place all cell phones, smart watches, electronic devices, and unauthorized study materials in your bag under your desk. If it is left in your pocket, it may be an academic offence.
- If you are feeling ill and unable to finish your exam, please bring it to the attention of an Exam Facilitator so it can be recorded before leaving the exam hall.
- This final examination consists of 8 question on 16 page (including this one), printed on both sides of the paper. *When you receive the signal to start, please make sure that your copy of the examination is complete.*
- Answer each question directly on the examination paper. If you need more space for a solution, use a "blank" page and *indicate clearly the part of your work that should be marked.*
- *To pass the course, you must achieve a grade of at least **40%** on this final examination.*
- As a student, you help create a fair and inclusive writing environment. If you possess an unauthorized aid during an exam, you may be charged with an academic offence.

Marking Guide

Nº 1:	_____/16
Nº 2:	_____/ 5
Nº 3:	_____/ 6
Nº 4:	_____/ 4
Nº 5:	_____/ 5
Nº 6:	_____/ 4
Nº 7:	_____/10
Nº 8:	_____/ 6
TOTAL:	_____/56

STUDENTS MUST HAND IN ALL EXAMINATION MATERIALS AT THE END

Question 1. Short Questions [16 marks]

Part (a) Naive Bayes [4 marks]

Which of the following is true about the Naive Bayes algorithm?
Circle **all** the correct answers.

(A) It works well with a small amount of labelled training data.

(B) It assumes a linear relationship between the features and the target variable.

(C) It assumes the features are independent.

(D) It can only handle binary classification problems.

Part (b) Gaussian Discriminant Analysis [4 marks]

Which of the following is true about Gaussian Discriminant Analysis (GDA)?
Circle **all** the correct answers.

(A) It is primarily used for unsupervised learning tasks.

(B) It works with continuous features.

(C) It assumes that the covariance matrices are the same for all classes.

(D) It does not make assumptions about the underlying data distribution.

Part (c) PCA [4 marks]

What is the primary objective of Principal Component Analysis (PCA)?
Circle **all** the correct answers.

(A) To minimize the reconstruction error.

(B) To maximize the variance of the reconstructions.

(C) To remove irrelevant data points.

(D) To remove irrelevant features.

Question 1. (continued)

Part (d) Towards Logistic Regression [2 marks]

When designing logistic regression, we considered the following model:

$$z = \mathbf{w}^{\mathbf{T}}\mathbf{x}, \qquad y = \sigma(z) = \frac{1}{1+e^{-z}}, \qquad \mathcal{L}(y,t) = \frac{1}{2}(y-t)^2$$

Describe one problem with this model. Make sure that you use a specific example.

Part (e) MAP versus MLE [2 marks]

Describe one advantage of Maximum-A-Posteriori (MAP) Estimation over Maximum Likelihood Estimation (MLE). Describe the advantage in at most 2 sentences.

Question 2. Linear Regression [5 marks]

Consider a modified linear regression model. The cost function is given below. You can interpret each r_j as a value we use to scale feature j.

$$\mathcal{E}(\mathbf{w}) = \frac{1}{2N} \sum_{i=1}^{N} \left(\left(\sum_{j=0}^{D} w_j r_j x_j^{(i)} \right) - t^{(i)} \right)^2$$

Derive the vectorized gradient $\nabla_{\mathbf{w}} \mathcal{E}(\mathbf{w})$. Show all your work.

You may use these quantities in your expression: $\mathbf{w}$ is the $(D+1) \times 1$ vector of weights. $\mathbf{r}$ is a $(D+1) \times 1$ vector. $\mathbf{R}$ is a $(D+1) \times (D+1)$ diagonal matrix where the diagonal entries are the values in $\mathbf{r}$. $\mathbf{X}$ is the $N \times (D+1)$ data matrix. $\mathbf{t}$ is the $N \times 1$ vector of target values.

Make sure that you show an expression of the non-vectorized gradient.

Put your final answer below.

Use the space on this "blank" page for scratch work, or for any solution that did not fit elsewhere.
Clearly label each such solution with the appropriate question and part number.

Question 3. Design a Small Neural Network [6 marks]

x_1	x_2	x_3	t
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

Table 1: Data Set for Neural Network

Design a **two-layer** neural network to represent the function in Table 1.
Every unit in the neural network should use the threshold activation function below.

$$\phi(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

The computations of the neural network are given below.

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \qquad \mathbf{h} = \phi(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}) \qquad y = \phi(\mathbf{W}^{(2)}\mathbf{h} + \mathbf{b}^{(2)})$$

Write your answers below. **You do not need to show any derivation.**

$\mathbf{W}^{(1)} =$ $\mathbf{b}^{(1)} =$

$\mathbf{W}^{(2)} =$ $\mathbf{b}^{(2)} =$

Explain what each hidden/output unit represents.

- The first hidden unit represents:
- The second hidden unit represents:
- The output unit represents:

Question 4. Naive Bayes and MAP [4 marks]

We will learn a **Naive Bayes** model to detect spam emails using the data set described below. Each data point is an email labelled spam or non-spam. The data set has **30 spam emails** and **60 non-spam emails**. We will consider two features corresponding to the words **spring** and **prize**. For each word, the feature has a value of 1 if the word appears in the email and 0 otherwise.

The table below provides the number of emails with a set of feature values and a given label. For example, the third row says there are 12 emails where **spring** does not appear in the email, **prize** appears in the email, and the email is labelled spam.

"spring" appears?	"prize" appears?	label	count
yes	yes	spam	8
yes	no	spam	3
no	yes	spam	12
no	no	spam	7
yes	yes	non-spam	11
yes	no	non-spam	17
no	yes	non-spam	4
no	no	non-spam	28

Table 2: Naive Bayes for Spam Detection

We will estimate the model parameters using MAP estimation with a Beta prior where $\mathbf{a} = \mathbf{3}, \mathbf{b} = \mathbf{4}$.

Part (a) Learn the Prior Probability [2 marks]

Determine the MAP estimate of $\boldsymbol{\pi}$, the prior probability that an email is spam.
Write your answer as a fraction of two integers.

Part (b) Learn the Conditional Probability [2 marks]

Determine the MAP estimate of $\boldsymbol{\theta_{11}}$ — the conditional probability that "prize" appears in the email given that the email is spam. **Write your answer as a fraction of two integers.**

Question 5. Maximum Likelihood Estimation for Gaussian Distributions [5 marks]

Suppose we want to learn the parameters of a **univariate** Gaussian distribution from data using **maximum likelihood estimation**. μ and σ^2 are the mean and variance of the Gaussian distribution, respectively. The PDF of the Gaussian distribution is given below.

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

where exp denotes the exponential function.
Assume we have N data points where $x^{(i)}$ denotes the feature value for the ith data point.

Derive the maximum likelihood estimate for σ^2, the variance of the Gaussian distribution.
Show all your work. Make sure you write down an expression for the log-likelihood of the N data points given μ and σ^2.

Put your final answer below.

Use the space on this "blank" page for scratch work, or for any solution that did not fit elsewhere.
Clearly label each such solution with the appropriate question and part number.

Question 6. Interpreting the Covariance Matrix [4 marks]

Assume we have a data set with 3 features x_1, x_2, x_3. We have provided the empirical covariance matrix and the result of performing spectral decomposition on the covariance matrix below.

$$\mathbf{\Sigma} = \begin{pmatrix} 1.21 & -1.55 & 0.48 \\ -1.55 & 5.63 & -0.04 \\ 0.48 & -0.04 & 6.07 \end{pmatrix}, \quad \mathbf{\Sigma} = \mathbf{Q\Lambda Q^T}, \quad \mathbf{Q} = \begin{pmatrix} -0.95 & -0.27 & -0.15 \\ -0.30 & 0.67 & 0.68 \\ 0.08 & -0.69 & 0.72 \end{pmatrix}, \quad \mathbf{\Lambda} = \begin{pmatrix} 0.69 & 0 & 0 \\ 0 & 6.29 & 0 \\ 0 & 0 & 5.93 \end{pmatrix}$$

Suppose that we learned a multi-variate Gaussian distribution using the data set above.

Recall that the covariance matrix acts on a vector by transforming it in an alternative coordinate system.

Complete the two sentences below.

Make sure each complete sentence includes the unit vector, has one of stretches and shrinks circled, and a value of the factor.

Sentence 1: In the direction of the unit vector $\begin{pmatrix} _____ \\ _____ \\ _____ \end{pmatrix}$, $\mathbf{\Sigma}$ stretches/shrinks (**circle one**) a vector by a factor of 0.69.

Sentence 2: In the direction of the unit vector $\begin{pmatrix} -0.15 \\ 0.68 \\ 0.72 \end{pmatrix}$, $\mathbf{\Sigma}$ stretches/shrinks (**circle one**) a vector by a factor of _____.

Question 7. GMM and EM [10 marks]

Consider the data set below with two features x_1 and x_2. We will learn a Gaussian Mixture Model (GMM) with 2 clusters ($K = 2$) using the Expectation-Maximization algorithm.

In the table below, i is the index of each data point, and $\mathbf{r_1^{(i)}}$ is the probability that data point i belongs to cluster 1.

i	x_1	x_2	$r_1^{(i)}$
1	1	2	0.9
2	2	7	0.8
3	5	4	0.15
4	6	3	0.05

Table 3: Data Set for GMM

The current cluster parameters are provided below. $\boldsymbol{\pi_1}$ is the mixing weight for cluster 1.
$\boldsymbol{\mu_1}, \boldsymbol{\Sigma_1}$ and $\boldsymbol{\mu_2}, \boldsymbol{\Sigma_2}$ are the mean vectors and the covariance matrices for clusters 1 and 2, respectively.

$$\boldsymbol{\pi_1} = 0.4, \quad \boldsymbol{\mu_1} = \begin{pmatrix} 1 & 5 \end{pmatrix}^T, \quad \boldsymbol{\Sigma_1} = \begin{pmatrix} 1 & 0.7 \\ 0.7 & 1 \end{pmatrix}, \quad \boldsymbol{\mu_2} = \begin{pmatrix} 5 & 2 \end{pmatrix}^T, \quad \boldsymbol{\Sigma_2} = \begin{pmatrix} 1 & -0.6 \\ -0.6 & 1 \end{pmatrix}$$

Part (a) E-Step [4 marks]

Starting with the provided information above, write down the **formulas** for updating $\boldsymbol{r_1^{(3)}}$ as part of the **E-step**. **Plug in all the numbers.**

You can write the probability of a data point in a Gaussian distribution as $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$ as long as you replace $\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\Sigma}$ by vectors/matrices with numbers.

Write the formulas for updating $\boldsymbol{r_1^{(3)}}$ below.

Question 7. (continued)

For your convenience, we have provided copies of the provided information on the previous page below.
In the table below, i is the index of each data point, and $\mathbf{r_1^{(i)}}$ is the probability that data point i belongs to cluster 1.

i	x_1	x_2	$r_1^{(i)}$
1	1	2	0.9
2	2	7	0.8
3	5	4	0.15
4	6	3	0.05

Table 4: A Copy of the Data Set for GMM

The current cluster parameters are provided below. $\boldsymbol{\pi_1}$ is the mixing weight for cluster 1.
$\boldsymbol{\mu_1}, \boldsymbol{\Sigma_1}$ and $\boldsymbol{\mu_2}, \boldsymbol{\Sigma_2}$ are the mean vectors and the covariance matrices for clusters 1 and 2, respectively.

$$\boldsymbol{\pi_1} = 0.4, \quad \boldsymbol{\mu_1} = \begin{pmatrix} 1 & 5 \end{pmatrix}^T, \quad \boldsymbol{\Sigma_1} = \begin{pmatrix} 1 & 0.7 \\ 0.7 & 1 \end{pmatrix}, \quad \boldsymbol{\mu_2} = \begin{pmatrix} 5 & 2 \end{pmatrix}^T, \quad \boldsymbol{\Sigma_2} = \begin{pmatrix} 1 & -0.6 \\ -0.6 & 1 \end{pmatrix}$$

Part (b) M-Step [6 marks]

Starting with the provided information above and assuming that we did NOT perform the E-step in part (a), write down the **formulas** for updating $\boldsymbol{\pi_1}$, $\boldsymbol{\mu_2}$ and $\boldsymbol{\Sigma_1}$ as part of the **M-step**. **Plug in all the numbers.**

Write the formula for updating $\boldsymbol{\pi_1}$ below.

Write the formula for updating $\boldsymbol{\mu_2}$ below.

Write the formula for updating $\boldsymbol{\Sigma_1}$ below. You can use $\boldsymbol{\mu_1}$ in your formula.

Question 8. Principal Component Analysis [6 marks]

Assume we have a data set with 3 features x_1, x_2, x_3. We have provided the empirical covariance matrix and the result of performing spectral decomposition on the covariance matrix below.

$$\mathbf{\Sigma} = \begin{pmatrix} 1.21 & -1.55 & 0.48 \\ -1.55 & 5.63 & -0.04 \\ 0.48 & -0.04 & 6.07 \end{pmatrix}, \quad \mathbf{\Sigma} = \mathbf{Q\Lambda Q^T}, \quad \mathbf{Q} = \begin{pmatrix} -0.95 & -0.27 & -0.15 \\ -0.30 & 0.67 & 0.68 \\ 0.08 & -0.69 & 0.72 \end{pmatrix}, \quad \mathbf{\Lambda} = \begin{pmatrix} 0.69 & 0 & 0 \\ 0 & 6.29 & 0 \\ 0 & 0 & 5.93 \end{pmatrix}$$

Part (a) First Principal Component [2 marks]

Suppose that we performed Principal Component Analysis on the data set above. Write down the first principal component. Explain why this is the first principal component in one sentence.

Write the **first principal component** below.

Explain **why** this is the first principal component in one sentence.

Part (b) Problem with PCA [2 marks]

What is one problem with the PCA analysis we performed above?
State the problem and explain why it is a problem in at most 2 sentences.

Question 8. (continued)

Part (c) Projecting onto a Subspace [2 marks]

In this part, we will project a vector onto the subspace spanned by the first two principal components from PCA. For simplicity, let's assume that the first two principal components from PCA are given below.

$$\mathbf{u_1} = \begin{pmatrix} u_{11} & u_{12} & u_{13} \end{pmatrix}^T, \mathbf{u_2} = \begin{pmatrix} u_{21} & u_{22} & u_{23} \end{pmatrix}^T.$$

Consider the subspace $\mathbf{S}$ where the basis vectors are $\mathbf{u_1}$ and $\mathbf{u_2}$ and its original is $\boldsymbol{\mu} = \begin{pmatrix} \mu_1 & \mu_2 & \mu_3 \end{pmatrix}^T$. Project the vector $\mathbf{v} = \begin{pmatrix} v_1 & v_2 & v_3 \end{pmatrix}^T$ onto the subspace $\mathbf{S}$. Let $\mathbf{z} = \begin{pmatrix} z_1 & z_2 \end{pmatrix}^T$ be the code vector and $\mathbf{w} = \begin{pmatrix} w_1 & w_2 & w_3 \end{pmatrix}^T$ be the reconstruction of $\mathbf{v}$.

Write down the formulas for computing z_2 and w_3.

Write down the formula for computing $\mathbf{z_2}$ below.

Write down the formula for computing $\mathbf{w_3}$ below. You can use $\mathbf{z_1}$ and $\mathbf{z_2}$ in the formula.

Use the space on this "blank" page for scratch work, or for any solution that did not fit elsewhere.
Clearly label each such solution with the appropriate question and part number.

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